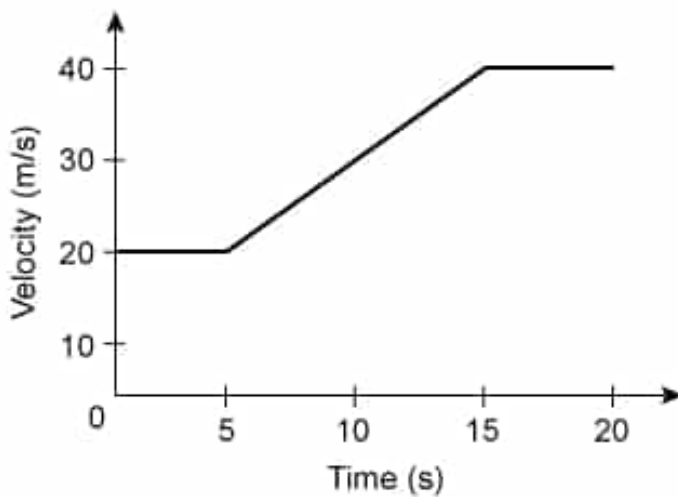


A car moves in a straight line along a paved road. The velocity of the car versus time is shown:



What is the displacement of the car from 0 s to 15 s?

- ☐ A. 300 m [18%]
- ✓ ☒ B. 400 m [59%]
- ☐ C. 500 m [9%]
- ☐ D. 600 m [13%]

Correct

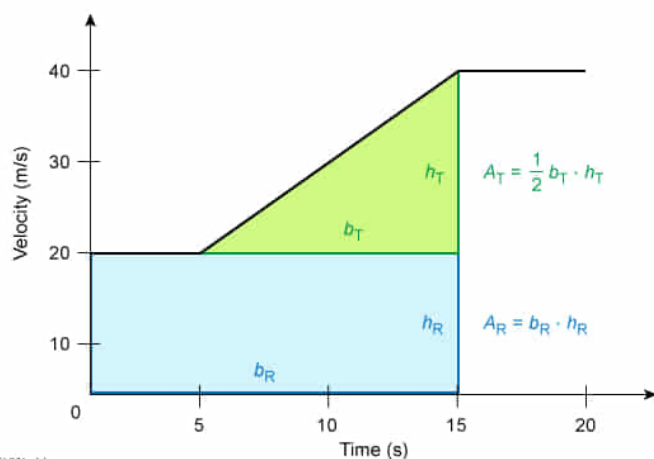
59% answered correctly

Explanation:

Displacement from area under the curve (velocity vs. time)

$$\Delta X = X_f - X_i = v \cdot t$$
$$\Delta X = \text{area under the curve}$$
$$\Delta X = A_T + A_R$$

- X_i Initial position
- X_f Final position
- v Velocity
- t Time



Although the relationships between translational variables can be expressed through **several equations**, graphs may also be used to explore these relationships visually.

For example, a graph of velocity v versus time t plots the velocity of an object as a function of time. The **area under the curve** of a velocity (y -axis) vs. time (x -axis) graph is equal to the **displacement** of an object over the corresponding period of time. Provided that the plot of the velocity vs. time graph remains above the x -axis, this displacement is equal to the total distance traveled. The equation for displacement ΔX of the object, in which X_i is the initial position and X_f is the final position, is given as:

$$\Delta X = (X_f - X_i) = v \cdot t$$

The **$v \cdot t$ component** of the equation shows that the displacement ΔX of a moving object is equal to the product of both axes (velocity and time) on the graph, which is **equivalent to the area under the curve** of this graph.

In this scenario, the given graph shows that the car initially moved at a constant velocity (flat slope segment), then accelerated at a constant rate (positive slope segment), after which it maintained a new constant velocity. The area under the curve can be calculated by visualizing the area into simple geometric shapes, in this case through the form of a rectangle and a triangle. Accordingly, adding the areas of these two shapes yields total displacement:

$$\begin{aligned} \text{Displacement} &= (\text{rectangle area}) + (\text{triangle area}) \\ \Delta X &= b_R h_R + \frac{1}{2} b_T h_T \\ &= (15 \text{ s} \cdot 20 \text{ m/s}) + \left(\frac{1}{2} \cdot 10 \text{ s} \cdot 20 \text{ m/s} \right) \\ &= 300 \text{ m} + 100 \text{ m} \\ &= 400 \text{ m} \end{aligned}$$

Based on these calculations, it can be concluded that the **car traveled 400 m (Choice B)** in the first 15 s of the journey.

Educational objective:

In a graph of velocity versus time, the displacement of a moving object over an interval of time can be derived from the graph by calculating the total area under the curve during that interval.

Time spent:0 sec

Subject:

Physics

QID:402034

System:

Mechanics and Energy